Lesson 28: Describing Statistical Relationships

English Stage 4 (Psychology) · Unit 4: Statistics in Research Reports

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 12.2 Describing Statistical Relationships.

In this lesson

Most research questions in psychology are about relationships between variables. You will learn to describe differences between groups (means, standard deviations, Cohen's *d*) and correlations between quantitative variables (Pearson's *r*), and you will practise the English of comparison and correlation, including "the more ..., the more ...".

Learning goals

After this lesson you should be able to:

- Describe a group difference with means, standard deviations and Cohen's *d*, and judge its size.
- Explain why an "effect size" does not prove cause and effect.
- Compute Pearson's *r* from *z* scores, interpret it, and recognize when it misleads.
- Write comparisons and correlations in clear academic sentences.

Key language

Term or phrase	Meaning	Example
statistical relationship	a difference in the average score on one variable across levels of another variable	Mean fear ratings differed across the three conditions.
effect size	the strength of a statistical relationship	an "effect size" of $d = 0.35$
Cohen's *d*	a measure of effect size for a difference between two groups or conditions	a large difference ($d = 0.82$)
bar graph	a graph used to show differences between the mean scores of two or more groups	the heights of the bars represent the means
positive / negative relationship	higher scores on one variable go with higher / lower scores on the other	self-esteem a week apart / name position and response time
linear / nonlinear relationship	points fit a straight line / a curved line	sleep and depression: an upside-down "U"
line graph / scatterplot	a point for the mean of *Y* at each level of *X* / a point for each individual	four quartiles of names / 25 students' self-esteem scores
Pearson's *r*	a measure of the strength of the correlation between two quantitative variables (−1.00 to +1.00)	$r = -.77$

Term or phrase	Meaning	Example
restriction of range	a limited range of scores on *X* or *Y*, which makes a relationship look weaker than it is	$r = 0$ for 18- to 24-year-olds only
the further ..., the faster ...	"the + comparative ..., the + comparative" for a correlation	The further toward the end of the alphabet students' last names were, the faster they tended to respond.
tend to be associated with	are linked with (no claim of cause)	Higher scores on one variable tend to be associated with higher scores on the other.

Rules and explanations

The ideas

Differences between groups. Report each group's mean and standard deviation, and show the means in a bar graph. **Ollendick et al. (2009)** randomly assigned children with an intense fear (for example, of dogs) to one-session exposure (facing the feared object with a therapist), education (learning about phobias and coping strategies) or a wait-list control. A clinician who did not know the condition rated severity from 1 to 8: exposure $M = 3.47$ ($SD = 1.77$), education $M = 4.83$ ($SD = 1.52$), control $M = 5.56$ ($SD = 1.21$). Both treatments worked, but exposure worked better.

Cohen's *d* = $(M1 - M2) / SD$

- $M1$ is usually the treatment group (or the larger mean); *d* is reported as positive.
- *SD* is usually the **pooled within-groups SD**: √[(sum of squared differences in Group 1 + sum in Group 2) / ($n1 + n2$)]. With equal group sizes, this is about $\sqrt{[(SD1^2 + SD2^2) / 2]}$. Informally, either group's *SD* can be used.
- *d* is a difference in *SD* units, like a *z* score: $d = 0.50$ means the means differ by half a standard deviation. **Cohen's guidelines:** about 0.20 small, 0.50 medium, 0.80 large. Exposure versus education: $d = 1.36 / \sqrt{[(1.77^2 + 1.52^2) / 2]} = 1.36 / 1.65 = 0.82$, a large difference.
- *d* means the same on any scale, so results can be compared and combined across studies.
- **Caution:** "effect size" sounds causal. If exercisers are happier than non-exercisers ($d = 0.35$), only an experiment shows that exercise *caused* it; a cross-sectional study shows only that exercisers *were* happier, by a small to medium amount.
- **Hyde (2007)** expressed sex differences as *d* (positive = men higher): mathematical problem solving +0.08, reading comprehension −0.09, smiling −0.40, aggression +0.50, attitudes toward casual sex +0.81, leadership effectiveness −0.02. Most differences are small, many below 0.10 ("trivial"); hence her "gender similarities hypothesis".
- **PSYC 110:** in Lecture 14 (What Motivates Us: Sex), Paul Bloom says that males are more physically violent "on average", which does not mean "you can't find violent women or non-violent men": an average difference like Hyde's +0.50. He also reports Larry Summers's claim that men and women have the same average aptitude but men show more *variation*: a claim about standard deviations, not means.

Correlations between quantitative variables

- **Carlson and Conard (2011)** offered MBA students free basketball tickets by email: the later in the alphabet their last names were, the faster they responded (a **negative** relationship between name position and response time). A **line graph** shows the mean response time for each quartile of names; a **scatterplot** of 25 students' self-esteem scores one week apart shows a **positive** relationship. Use a line graph when X has few values, a scatterplot when it has many.
- **Pearson's *r*:** −1.00 to +1.00, with 0 meaning no relationship. **Cohen's guidelines:** about ±.10 small, ±.30 medium, ±.50 large. The sign shows direction, not strength.
- **Computing *r*:** turn every score into a z score, multiply each person's two z scores (the cross-product) and take the mean: $r = \Sigma(zxzy) / N$. Book example: X = 4, 7, 2, 5, 2 (M = 4.00, SD = 1.90); Y = 30, 54, 23, 43, 50 (M = 40.00, SD = 11.78); r = 2.64 / 5 = .53. (The book prints 0.26 as the z score of 43; it is 0.25, which does not change r.)
- **Pearson's *r* misleads** when the relationship is **nonlinear** (sleep and depression: people who sleep about eight hours are least depressed, yet r would be near zero), so make a scatterplot first; and with **restriction of range** (age and enjoyment of hip hop: r = −.77 overall, but 0 for 18- to 24-year-olds only).

Language focus

1. Comparing groups: give the direction, then the size.

- "Both treatments worked, **but** the exposure treatment worked **better than** the education treatment."
- "There was **a large difference** (d = 0.82) **between** the exposure and education conditions."

2. Describing correlations: "**the + comparative …, the + comparative …**" with *tend to* (a tendency, not a rule).

- "**The higher** students scored on the first occasion, **the higher** they **tended to** score on the second occasion."
- For correlational or cross-sectional data, write "is associated with", not "causes".
- APA numbers: d = 0.82 has a zero before the point (d can exceed 1); r = .53 has none (r cannot exceed 1), as in the book's "r = −.77".

> **Note for Arabic speakers** - In "the + comparative" sentences, both halves need *the* and a comparative, followed by subject and verb: "The more hours students slept, the more pairs they recalled" (not "More students slept, more they recalled").

Practice

Part 1: Reading questions

1. In Ollendick et al.'s study, what were the three conditions, and who rated each child's phobia?
2. What does a Cohen's d of 1.20 mean in words?
3. Why can the term "effect size" be misleading?

4. On which of Hyde's variables do men and women differ most, and what is the "gender similarities hypothesis"?
5. When do researchers use a line graph rather than a scatterplot?
6. Are *r* values of +.30 and −.30 equally strong? Explain.
7. In what two situations can Pearson's *r* be misleading?
8. Compute *d* for exposure versus the wait-list control in Ollendick et al.'s study, using $\sqrt{[(SD1^2 + SD2^2) / 2]}$.

Part 2: Exercises from the book (chapter exercises, section 12.5)

Exercise 4. Practice: The following data represent scores on the Rosenberg Self-Esteem Scale for a sample of 10 Japanese university students and 10 American university students. Compute the means and standard deviations of the two groups, make a bar graph, compute Cohen's *d*, and describe the strength of the relationship in words. Japan: 25, 20, 24, 28, 30, 32, 21, 24, 20, 26. United States: 27, 30, 34, 37, 26, 24, 28, 35, 33, 36.

Exercise 5. Practice: The hypothetical data that follow are extraversion scores and the number of Facebook friends for 15 university students. Make a scatterplot for these data, compute Pearson's *r*, and describe the relationship in words. (Extraversion, friends): (8, 75), (10, 315), (4, 28), (6, 214), (12, 176), (14, 95), (10, 120), (11, 150), (4, 32), (13, 250), (5, 99), (7, 136), (8, 185), (11, 88), (10, 144).

Part 3: Language exercise

- A. Write a "the + comparative" sentence for (1) age and enjoyment of hip hop ($r = -.77$); (2) extraversion and Facebook friends.
- B. Correct this overstatement about a cross-sectional study: "Exercise causes happiness ($d = 0.35$)."
- C. Describe in words Hyde's results for smiling (−0.40) and attitudes toward casual sex (+0.81).

Speak with your tutor

- "Let me explain Cohen's *d* in my own words with Ollendick et al.'s phobia study. Then ask me what $d = 0.20$ would mean in practice."
- "Help me rewrite these claims to match the evidence: 'Exercise makes people happier'; 'Late-alphabet names cause fast responses'."
- "In PSYC 110 Lecture 14, Paul Bloom discusses sex differences in aggression and Summers's claim about variation. Let me explain the difference between a difference in means and a difference in standard deviations, and connect it to Hyde."
- "Ask me why a study of sleep and depression could find $r = 0$ although the variables are strongly related."

Writing task

Your research project: results, part 2 (describing the relationships). Use the Lesson 27 dataset (**invented data for practice**: 20 students randomly assigned to a nap or a wake condition; recall out of 40):

Data (participants 1 to 10 in order)	
Nap: recall	24, 19, 22, 26, 21, 23, 18, 21, 25, 21
Nap: minutes asleep	70, 60, 80, 85, 55, 70, 75, 65, 70, 70
Wake: recall	18, 15, 21, 16, 12, 19, 16, 22, 14, 17

Write 100 to 150 words that (1) compare the conditions with means, standard deviations and Cohen's *d* (the sums of squared differences are 58 for nap and 86 for wake), and (2) describe the correlation between minutes asleep and recall in the nap condition (Pearson's *r*). Make a scatterplot first; comment on linearity and range. Describe each size in words. If you work on your own topic, use your own invented dataset and ask your tutor to check your *d* and *r*.

Checklist:

- Did you state which condition recalled more before giving the size of the difference?
- Are *d* and *r* given to two decimal places, with no zero before the point in *r*?
- Did you describe each size in words (small, medium, large)?
- Did you use "the + comparative ..., the + comparative" correctly?
- Did you avoid causal language for the correlation?
- Did you mention linearity and the range of minutes asleep?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Answer key

Part 1

1. Exposure, education and wait-list control; a clinician who did not know the child's treatment, on a 1-to-8 scale.
2. The two means differ by 1.20 standard deviations: a very large difference in psychological research.
3. It suggests that the difference is caused by being in one group, but only an experiment with random assignment shows that.
4. Attitudes toward casual sex (+0.81). Most sex differences are small or trivial, so men and women can be seen as fundamentally similar.
5. When the *x*-axis variable has (or is organized into) a small number of distinct values, such as four quartiles.
6. Yes; the sign shows only the direction of the relationship.
7. When the relationship is nonlinear, and when a variable has a restricted range.
8. $\sqrt{[(1.77^2 + 1.21^2) / 2]} = 1.52$; $d = (5.56 - 3.47) / 1.52 = 1.38$, a very large difference.

Part 2

- Exercise 4. Japan: $M = 25.00$, $SD = 3.90$ (dividing by N) or 4.11 ($N - 1$). United States: $M = 31.00$, $SD = 4.36$ or 4.59. Bar graph: bars of 25 and 31 on a *y*-axis starting at 0. Pooled *SD* (book's formula) = $\sqrt{[(152 + 190) / 20]} = 4.14$; $d = 6.00 / 4.14 = 1.45$ (with $N - 1$: 1.38). (example answer) The American students scored about one and a half standard deviations higher: a very large difference, well above 0.80.
- Exercise 5. $r = .42$. Scatterplot: extraversion on the *x*-axis, friends on the *y*-axis; the points rise from lower left to upper right with much scatter, and one student (10, 315) lies far above the rest. (example answer) There was a medium-to-strong positive correlation: more extraverted students tended to have more Facebook friends.

Part 3 (example answers)

- A1. The older people were, the less they tended to enjoy hip hop music.
- A2. The more extraverted students were, the more Facebook friends they tended to have.
- B. Exercisers were happier than non-exercisers by a small to medium amount ($d = 0.35$).
- C. Women smiled more than men, a difference close to medium size ($d = -0.40$). Men scored higher than women on attitudes toward casual sex, a large difference ($d = +0.81$).

Writing task. Pooled $SD = \sqrt{[(58 + 86) / 20]} = 2.68$; $d = (22.00 - 17.00) / 2.68 = 1.86$ (with $N - 1$: $SD = 2.83$, $d = 1.77$). $r = .45$. Example: "Participants in the nap condition recalled more word pairs ($M = 22.00$, $SD = 2.54$) than those in the wake condition ($M = 17.00$, $SD = 3.09$), a very large difference ($d = 1.86$). In the nap condition, minutes asleep and recall showed a medium-to-strong positive correlation ($r = .45$): the longer participants slept, the more pairs they tended to recall. The scatterplot showed a roughly linear trend with considerable scatter, but the narrow range of sleep (55 to 85 minutes) may have restricted the correlation."

My notes